

ALGORITHM ARENA

[Problem Title]

Match Protocol and Tournament Evaluation

DOCUMENT B: ARENA AND EVALUATION SPECIFICATION

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Standard Document B template specifying controller protocols, match loops, tournament rules, baseline algorithms, and scoring systems.

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1 Overview

Define the matchmaking scope, controller execution model, and evaluation goals.

2 Controller Protocol

2.1 Lifecycle Methods

- `initialize()`: Process setup and static resource loading.
- `reset()`: Memory cleanup before each match.
- `step(observation)`: Synchronous action dispatch.

2.2 Observation and Action Contracts

Data types and ranges for input and output.

3 Match Protocol

3.1 Horizon and Termination

Step bounds T and early stopping conditions.

3.2 Error Handling and Fallbacks

Behavior on timeouts, unhandled exceptions, and invalid actions.

4 Randomness and Reproducibility

4.1 Seed Hierarchy

Hierarchical derivation from `Seedmatch` to environment and controller PRNGs.

4.2 Determinism Contract

Given the same seed and code, output must be 100% reproducible.

5 Tournament Format and Scoring

5.1 Matchmaking Format

Round-robin, Swiss system, or benchmark ladder.

5.2 Ranking Metrics and Tie-Breaking

Primary and secondary leaderboard criteria.

6 Benchmark Baseline Suite

Descriptions of reference baseline opponents (Naive, Standard, Strong).